Section D3 :

1. Name of the Medicinal Product
   OZPROSIN ™ (Omeprazole for Injection 40 mg/Vial)

2. Qualitative and Quantitative Composition
   Each lyophilized 10 ml vial contains: Omeprazole sodium, equivalent to 40 mg Omeprazole

3. Pharmaceutical Form
   Parenteral: Powder for Injection (Lyophilised Cake)

4. Product Description:
   Omeprazole for injection is a white to almost white lyophilized powder or cake yields a clear colourless to pale yellow colour solution on reconstitution (with 10 mL of water) intended to be used only for intravenous injection. It has a molecular weight 385.41.

5. Pharmacodynamics:
   Pharmacotherapeutic group: Omeprazole for Injection used as Proton Pump Inhibitors

   Mechanism of action and Pharmacodynamic effects:
   Omeprazole, a racemic mixture of two enantiomers reduces gastric acid secretion through a highly targeted mechanism of action. It is a specific inhibitor of the acid pump in the parietal cell. It is rapidly acting and provides control through reversible inhibition of gastric acid secretion with once-daily dosing. Omeprazole is a weak base and is concentrated and converted to the active form in the highly acidic environment of the intracellular canaliculi within the parietal cell, where it inhibits the enzyme H+, K+ATPase - the acid pump. This effect on the final step of the gastric acid formation process is dose-dependent and provides for highly effective inhibition of both basal acid secretion and stimulated acid secretion, irrespective of stimulus.

   Pharmacodynamic effects
   All Pharmacodynamic effects observed can be explained by the effect of Omeprazole on acid secretion.

   Effect on gastric acid secretion
   Intravenous Omeprazole produces a dose dependent inhibition of gastric acid secretion in humans. In order to immediately achieve a similar reduction of intragastric acidity as after repeated dosing with 20 mg orally, a first dose of 40 mg intravenously is recommended. This results in an immediate decrease in intragastric acidity and a mean decrease over 24 hours of 90% for both IV injection and IV infusion.
   The inhibition of acid secretion is related to the area under the plasma concentrationtime curve (AUC) of Omeprazole and not to the actual plasma

concentration at a given time. No tachyphylaxis has been observed during treatment with Omeprazole.

Effect on H. pylori

H. pylori is associated with peptic ulcer disease, including duodenal and gastric ulcer disease. H. pylori is a major factor in the development of gastritis. H. pylori together with gastric acid are major factors in the development of peptic ulcer disease. H. pylori is a major factor in the development of atrophic gastritis which is associated with an increased risk of developing gastric cancer. Eradication of H. pylori with Omeprazole and antimicrobials is associated with high rates of healing and long-term remission of peptic ulcers.

Other effects related to acid inhibition

During long-term treatment gastric glandular cysts have been reported in a increased frequency. These changes are a physiological consequence of pronounced inhibition of acid secretion, are benign and appear to be reversible. Decreased gastric acidity due to any means including proton pump inhibitors, increases gastric counts of bacteria normally present in the gastrointestinal tract. Treatment with acid-reducing medicinal products may lead to slightly increased risk of gastrointestinal infections such as Salmonella and

Campylobacter.

During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours. Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

6. Pharmacokinetics

Distribution

The apparent volume of distribution in healthy subjects is approximately 0.3 l/kg body weight. Omeprazole is 97% plasma protein bound.

Biotransformation

Omeprazole is completely metabolised by the cytochrome P450 system (CYP). The major part of its metabolism is dependent on the polymorphically expressed CYP2C19, responsible for the formation of hydroxyl Omeprazole, the major metabolite in plasma. The remaining part is dependent on another specific isoform, CYP3A4, responsible for the formation of Omeprazole sulphone. As a consequence of high affinity of Omeprazole

to CYP2C19, there is a potential for competitive inhibition and metabolic drug-drug interactions with other substrates for CYP2C19. However, due to low affinity to CYP3A4, Omeprazole has no potential to inhibit the metabolism of other CYP3A4 substrates. In addition, Omeprazole lacks an inhibitory effect on the main CYP enzymes.

Approximately 3% of the Caucasian population and 15–20% of Asian populations lack a functional CYP2C19 enzyme and are called poor metabolisers. In such individuals the metabolism of Omeprazole is probably mainly catalysed by CYP3A4. After repeated once-daily administration of 20 mg Omeprazole, the mean AUC was 5 to 10 times higher in poor metabolisers than in subjects having a functional CYP2C19 enzyme (extensive metabolisers). Mean peak plasma concentrations were also higher, by 3 to 5 times. These findings have no implications for the posology of Omeprazole.

Elimination

Total plasma clearance is about 30-40 l/h after a single dose. The plasma elimination half-life of Omeprazole is usually shorter than one hour both after single and repeated once-daily dosing. Omeprazole is completely eliminated from plasma between doses with no tendency for accumulation during once-daily administration. Almost 80% of a dose of Omeprazole is excreted as metabolites in the urine, the remainder in the faeces, primarily originating from bile secretion.

Linearity/non-linearity

The AUC of Omeprazole increases with repeated administration. This increase is dose-dependent and results in a non-linear dose-AUC relationship after repeated administration. This time- and dose-dependency is due to a decrease of first pass metabolism and systemic clearance probably caused by an inhibition of the CYP2C19 enzyme by Omeprazole and/or its metabolites (e.g. the sulphone). No metabolite has been found to have any effect on gastric acid secretion.

Special populations

Impaired hepatic function

The metabolism of Omeprazole in patients with liver dysfunction is impaired, resulting in an increased AUC. Omeprazole has not shown any tendency to accumulate with oncedaily dosing.

Impaired renal function

The pharmacokinetics of omeprazole, including systemic bioavailability and elimination rate, are unchanged in patients with reduced renal function.

Elderly

The metabolism rate of omeprazole is somewhat reduced in elderly subjects (75-79 years of age).

7. Indication

Omeprazole for Injection is gastric acid secretion inhibitor otherwise known as Proton Pump Inhibitor. It is indicated for stress ulceration, management of bleeding peptic ulcers. Protection against acid aspiration, duodenal ulcer, gastric ulcer and reflux oesophagitis. Also, in treatment of Zollinger-Ellison Syndrome and other acid related diseases where oral therapy is inappropriate.

8. Recommended Dose

Duodenal ulcer, gastric ulcer and reflux oesophagitis: patients who cannot be given oral medication can be treated parenterally with 40mg once daily. The usual treatment period before transfer to oral treatment is 2 - 3 days.

In Zollinger-Ellison Syndrome the dose should be adjusted accordingly to individual. Higher doses and /or several doses daily may be required.

Intravenous treatment can be given as injection, whereby the solution for injection must be given slowly over a period of at least 2 minutes and with a rate of maximum 4ml per minute.

Impaired renal function

A dose adjustment is not necessary for patients with impaired renal function.

Impaired liver function

In patients with impaired liver function clearance is greatly reduced.

Elderly patients

Dose adjustment is not necessary in elderly patients.

Children

There is only limited experience of treatment in children.

9. Route of Administration

Ozprosin (Omeprazole for Injection 40 mg/vial) is administered intravenously (I.V)

either as a bolus injection or as a continuous infusion.

10. Contraindication

Known hypersensitivity to Omeprazole, substituted benzimidazoles or to any of the excipients.
Omeprazole like other proton pump inhibitors (PPIs) should not be used concomitantly with nelfinavir.

11. Warning and Precautions:

In the presence of any alarm symptoms (e.g., significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis or melena) and when gastric ulcer is suspected or present, malignancy should be excluded, as treatment may alleviate symptoms and delay diagnosis.

Co-administration of atazanavir with proton pump inhibitors is not recommended .If the combination of atazanavir with a proton pump inhibitor is judged unavoidable, close clinical monitoring (e.g. virus load) is recommended in combination with an increase in the dose of atazanavir to 400 mg with 100 mg of ritonavir; Omeprazole 20 mg should not be exceeded.

Omeprazole, as with all acid-blocking medicines, may reduce the absorption of vitamin B12 (cyanocobalamin) due to hypo- or achlorhydria. This should be considered in patients with reduced body stores or risk factors for reduced vitamin B12 absorption on long-term therapy.

Omeprazole is a CYP2C19 inhibitor. When starting or ending treatment with Omeprazole, the potential for interactions with drugs metabolised through CYP2C19 should be considered. An interaction is observed between clopidogrel and Omeprazole. The clinical relevance of this interaction is uncertain. As a precaution, concomitant use of Omeprazole and clopidogrel should be discouraged.
Treatment with proton pump inhibitors may lead to slightly increased risk of gastrointestinal infections such as Salmonella and Campylobacter.

Subacute cutaneous lupus erythematosus (SCLE)

Proton pump inhibitors are associated with very infrequent cases of SCLE. If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Omeprazole. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia

Severe hypomagnesaemia has been reported in patients treated with PPIs like omeprazole for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur, but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPIs with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture

Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10–40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhoea

Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhoea, especially in hospitalized patients. This diagnosis should be considered for diarrhoea that does not improve.

Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency

Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Interference with laboratory tests

Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. To avoid this interference, Omeprazole treatment should be stopped for at least 5 days before CgA measurements. If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

As in all long-term treatments, especially when exceeding a treatment period of 1 year, patients should be kept under regular surveillance.

This medicinal product contains less than 1mmol sodium (23 mg) per dose, i.e. essentially “sodium free”.

12. Interaction with Other Medicaments

<u>Effects of Omeprazole on the pharmacokinetics of other active substances</u>

<u>Active substances with pH dependent absorption</u>

The decreased intragastric acidity during treatment with Omeprazole might increase or decrease the absorption of active substances with a gastric pH dependent absorption.

Nelfinavir, atazanavir

The plasma levels of nelfinavir and atazanavir are decreased in case of co-administration with Omeprazole.
Concomitant administration of Omeprazole with nelfinavir is contraindicated. Co-administration of Omeprazole (40 mg once daily) reduced mean nelvinavir exposure by ca. 40% and the mean exposure of the pharmacologically active metabolite M8 was reduced by ca. 75-90%. The interaction may also involve CYP2C19 inhibition.

Concomitant administration of Omeprazole with atazanavir is not recommended. Concomitant administration of Omeprazole (40 mg once daily) and atazanavir 300 mg/ritonavir 100 mg to healthy volunteers resulted in a 75% decrease of the atazanavir exposure. Increasing the atazanavir dose to 400 mg did not compensate for the impact of Omeprazole on atazanavir exposure. The co-administration of Omeprazole (20 mg once daily) with atazanavir 400 mg/ritonavir 100 mg to healthy volunteers resulted in a decrease of approximately 30% in the atazanavir exposure as compared to atazanavir 300 mg/ritonavir 100 mg once daily.

Digoxin

Concomitant treatment with Omeprazole (20 mg daily) and digoxin in healthy subjects increased the bioavailability of digoxin by 10%. Digoxin toxicity has been rarely reported. However, caution should be exercised when Omeprazole is given at high doses in elderly patients. Therapeutic drug monitoring of digoxin should be then be reinforced.

Clopidogrel

Results from studies in healthy subjects have shown a pharmacokinetic (PK)/pharmacodynamic (PD) interaction between clopidogrel (300 mg loading dose/75 mg daily maintenance dose) and Omeprazole (80 mg p.o. daily) resulting in a decreased exposure to the active metabolite of clopidogrel by an average of 46% and a decreased maximum inhibition of (ADP induced) platelet aggregation by an average of 16%.

Inconsistent data on the clinical implications of this PK/PD interaction in terms of major cardiovascular events have been reported from observational and clinical studies. As a precaution, concomitant use of Omeprazole and clopidogrel should be discouraged.

Other active substances

The absorption of posaconazole, erlotinib, ketoconazole and itraconazole is significantly reduced and thus clinical efficacy may be impaired. For posaconazole and erlotinib concomitant use should be avoided.

Active substances metabolised by CYP2C19

Omeprazole is a moderate inhibitor of CYP2C19, the major omeprazole metabolising enzyme. Thus, the metabolism of concomitant active substances also metabolised by CYP2C19, may be decreased and the systemic exposure to these substances increased. Examples of such drugs are R-warfarin and other vitamin K antagonists, cilostazol, diazepam and phenytoin.

Cilostazol

Omeprazole given in doses of 40 mg to healthy subjects in a cross-over study increased Cmax and AUC for cilostazol by 18% and 26% respectively, and one of its active metabolites by 29% and 69% respectively.

Phenytoin
Monitoring phenytoin plasma concentration is recommended during the first two weeks after initiating Omeprazole treatment and if a phenytoin dose adjustment is made, monitoring and a further dose adjustment should occur upon ending Omeprazole treatment.

Unknown mechanism
Saquinavir
Concomitant administration of Omeprazole with saquinavir/ritonavir resulted in increased plasma levels up to approximately 70% for saquinavir associated with good tolerability in HIV-infected patients.

Tacrolimus
Concomitant administration of omeprazole has been reported to increase the serum levels of tacrolimus. A reinforced monitoring of tacrolimus concentrations as well as renal function (creatinine clearance) should be performed, and dosage of tacrolimus adjusted if needed.

Methotrexate

When given together with proton pump inhibitors, methotrexate levels have been reported to increase in some patients. In high-dose methotrexate administration a temporary withdrawal of Omeprazole may need to be considered.

Effects of other active substances on the pharmacokinetics of omeprazole

Inhibitors of CYP2C19 and/or CYP3A4

Since Omeprazole is metabolised by CYP2C19 and CYP3A4, active substances known to inhibit CYP2C19 or CYP3A4 (such as clarithromycin and voriconazole) may lead to increased Omeprazole serum levels by decreasing omeprazole's rate of metabolism.

Concomitant voriconazole treatment resulted in more than doubling of the Omeprazole exposure. As high doses of Omeprazole have been well-tolerated adjustment of the Omeprazole dose is not generally required. However, dose adjustment should be considered in patients with severe hepatic impairment and if long-term treatment is indicated.

Inducers of CYP2C19 and/or CYP3A4

Active substances known to induce CYP2C19 or CYP3A4 or both (such as rifampicin and St John's wort) may lead to decreased Omeprazole serum levels by increasing omeprazole's rate of metabolism.

13. Pregnancy and lactation

Pregnancy

Results from three prospective epidemiological studies indicate no evidence of adverse events of Omeprazole on pregnancy or on the health of the foetus/new-born child. Omeprazole can be used during pregnancy.

Breastfeeding

Omeprazole is excreted in breast milk but is not likely to influence the child when therapeutic doses are used.

Fertility

Animal studies with the racemic mixture Omeprazole, given by oral administration do not indicate effects with respect to fertility.

14. Side Effects

Like all medicines, this medicine can cause side effects, although not everybody gets them. If you notice any of the following rare but serious side effects, stop using Omeprazole and contact a doctor immediately.

The most common side effects are headache, abdominal pain, constipation, diarrhoea, flatulence and nausea/vomiting.

SOC/Frequency	**Adverse reaction**
Blood and lymphatic system disorders	
Rare:	Leukopenia, thrombocytopenia
Very rare:	Agranulocytosis, pancytopenia
Immune system disorders	
Rare:	Hypersensitivity reactions e.g. fever, angioedema and anaphylactic reaction/shock
Metabolism and nutrition disorders	
Rare:	Hyponatraemia
Not known	Hypomagnesaemia; severe hypomagnesaemia may result in hypocalcaemia. Hypomagnesaemia may also be associated with hypokalaemia
Psychiatric disorders	
Uncommon	Insomnia
Rare:	Agitation, confusion, depression
Very rare:	Aggression, hallucinations
Nervous system disorders	
Common:	Headache
Uncommon:	Dizziness, paraesthesia, somnolence
Rare:	Taste disturbance
Eye disorders	
Rare:	Blurred vision
Ear and labyrinth disorders	
Uncommon:	Vertigo
Respiratory, thoracic and mediastinal disorders	
Rare:	Bronchospasm

SOC/Frequency	Adverse reaction
Gastrointestinal disorders	
Common:	Abdominal pain, constipation, diarrhoea, flatulence, nausea/vomiting
Rare:	Dry mouth, stomatitis, gastrointestinal candidiasis
Not known	Microscopic colitis
Hepatobiliary disorders	
Uncommon:	Increased liver enzymes
Rare:	Hepatitis with or without jaundice
Very rare:	Hepatic failure, encephalopathy in patients with pre-existing liver disease
Skin and subcutaneous tissue disorders	
Uncommon:	Dermatitis, pruritus, rash, urticaria
Rare:	Alopecia, photosensitivity
Very rare:	Erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (TEN)
Not known:	Subacute cutaneous lupus erythematosus
Musculoskeletal and connective tissue disorders	
Uncommon:	Fracture of the hip, wrist or spine
Rare:	Arthralgia, myalgia
Very rare:	Muscular weakness
Renal and urinary tract disorders	
Rare:	Interstitial nephritis
Reproductive system and breast disorders	
Very rare:	Gynaecomastia
General disorders and administration site conditions	
Uncommon:	Malaise, peripheral oedema
Rare:	Increased sweating

Irreversible visual impairment has been reported in isolated cases of critically ill patients who have received omeprazole intravenous injection, especially at high doses, but no causal relationship has been established.

<u>Subacute Cutaneous Lupus Erythematosus (SCLE)</u>

Skin and subcutaneous tissue disorders

Frequency 'not known': Subacute cutaneous lupus erythematosus

Interstitial Nephritis
Renal and urinary disorders: Interstitial nephritis
Hypomagnesaemia
Metabolism and nutritional disorders Frequency “not known”: hypomagnesaemia.
Clostridium Difficile Diarrhoea
Infections & infestations: Clostridium difficile associated diarrhoea.
Fracture
Musculoskeletal disorders
Frequency “uncommon”: Fracture of the hip, wrist or spine.
Fundic Gland Polyps (Benign)
Gastrointestinal disorders
Frequency “common”: Fundic gland polyps (benign)
Vitamin B12 Deficiency
Metabolic/Nutritional: Vitamin B12 deficiency

15. Symptoms and Treatment of Overdose

There is limited information available on the effects of overdoses of omeprazole in humans. In the literature, doses of up to 560 mg have been described and occasional reports have been received when single oral doses have reached up to 2,400 mg omeprazole (120 times the usual recommended clinical dose). Nausea, vomiting, dizziness, abdominal pain, diarrhoea and headache have been reported. Also, apathy, depression and confusion have been described in single cases.

The symptoms described have been transient and no serious outcome has been reported. The rate of elimination was unchanged (first order kinetics) with increased doses. Treatment, if needed, is symptomatic.

Intravenous doses of up to 270 mg on a single day and up to 650 mg over a three-day period have been given in clinical trials without any dose-related adverse reactions.

16. Effects on Ability to Drive and Use Machine

Omeprazole is not likely to affect the ability to drive or use machines. Adverse drug reactions such as dizziness and visual disturbances may occur. If affected, patients should not drive or operate machinery.

17. Preclinical Safety Data   Not – Applicable.

18. Instructions for use

Ozprosin (Omeprazole for Injection 40mg/vial) is a Prescription only Medicine. Ozprosin (Omeprazole for Injection 40mg/vial) is obtained by dissolving the lyophilized substance in the solvent (with 10 mL of water for injection). The stability of Omeprazole is influenced by the pH of the solution for injection, which is why no other solvents or quantities should be used for dilution. Use only clear, colourless or pale-yellow solutions. Do not use if any particles are present in the reconstituted solution. The reconstituted solution is for single use only.

Preparation

NOTE: Steps 1 to 5 must be performed in immediate sequence:

1. With a syringe draw all of the solvent from the Ampoule (10 mL).
2. Add approximately 5 mL of the solvent to the vial with lyophilized Omeprazole.
3. Withdraw as much air as possible from the vial back into the syringe. This will make it easier to add the remaining solvent.
4. Add the remaining solvent into the vial, make sure the syringe is empty.
5. Rotate and shake the vial to ensure all the lyophilized Omeprazole has dissolved.

Ozprosin (Omeprazole for Injection 40mg/vial) must be given only as an Intravenous

Injection and it must not be added to infusion solutions. After reconstitution the injection should be given slowly over a period of at least 2.5 minutes at a maximum rate of 4 mL per minute.

19. Storage Conditions

Ozprosin (Omeprazole for Injection 40mg/vial) should be store at a temperature not exceeding 30 °C. Protect from light

Ozprosin - Reconstituted Omeprazole for Injection is to be used immediately upon reconstitution.

20. Shelf Life

1. Shelf life in the medicinal product as packaged for sale: 24 months

2. Shelf life after reconstitution according to directions: To be used immediately upon reconstitution.

3. Shelf life after first opening the container: Not applicable

21. Nature and contents of the container

Pack Size: 1 Vial

The product is packed as (2 ml fill volume) 40 mg in 10 mL(Type I) Amber glass lyo vial {as per COA attached in P7} , stoppered with 13 mm ready to sterilize grey bromobutyl rubber stoppers and sealed with 13mm aluminium flip-off seals with magenta color button further labeled vials packed in outer carton with leaflet.

22. Manufactured By; Gland Pharma Limited
Survey No.:143 to 148, 150 & 151
Near Gandimaisamma Cross Road,
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